Ecole Supérieure d'Informatique et du Numérique
COLLEGE OF ENGINEERING & ARCHITECTURE

Introduction to Cybersecurity

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Iptables

- Perform packet filtering, mainly to ensure firewall functions: forbid, acepeter.
- Perform Network Address Translation (NAT) operations
- Perform packet marking operations, to apply special treatment to them

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Chains and rules

- A string is a stack of rules
- Each rule in a string is composed of two parts: (a criterion and a policy)
- The criterion specifies the cases of application
- The politician specifies the action accomplished
- If the criterion does not apply to the packet, the following rule is considered
- **==> the order of the rules is important**
- If no rules apply, the **policy associated with the channel is reviewed.**

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Tables et leurs chaînes

There are three tables that will be used to contain "filtering" rules:

- **FILTER: (INPUT, OUTPUT, FORWORD)**
- **NAT**: (PREROUTING, POSTROUTING, OUTPUT)
- **MANGLE**: (PREROUTING, OUTPUT)

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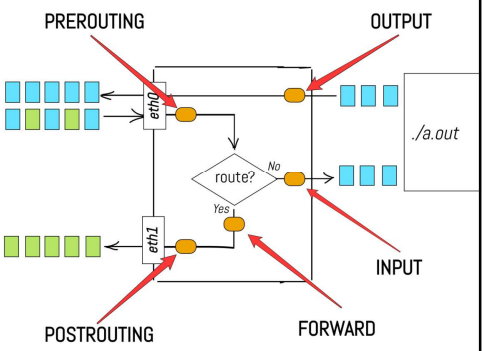
Les différentes chaines prédéfinies

- **INPUT:** Traffic for the local machine
- **OUTPUT:** the traffic going out o the machine to the network
- **FORWARD:** traffic that passes through the machine
- **PREROUTING:** This chain can modify a traffic as soon as it enters the system before it is routed
- **POSTROUTING:** this chain car modify a traffic but after routing



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the "Filter" table

- **The INPUT.**
  This chain will decide the fate of packets entering the host locally.
- **The OUTPUT chain.**
  Here, only the packets emitted by the local host will be filtered
- **The FORWARD.**
  Finally, packets that traverse the host, following the implanted routes, will be filtered here.

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La table NAT

- **The PREROUTING channel : Allows you to do destination address translation. This method is interesting if we want to make the outside world believe, for example, that there is a WEB server on port 80 of the gateway, when it is hosted by a host of the private network, on port 8080.**
- **The POSTROUTING : It allows you to do the translation of the address of the source, as well as address masking, the classic method to connect a private network as a client of the Internet, with a single "official" IP address.**
- **The OUTPUT chain: This will allow you to change the destination of packets generated locally (by the gateway itself).**

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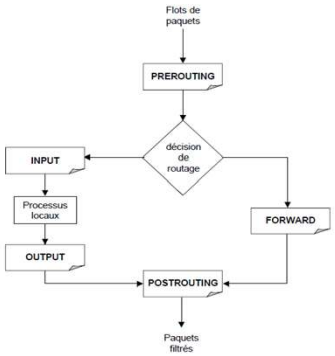
La table MANGLE

- This table allows the marking of incoming (**PREROUTING**) and locally generated (**OUTPUT**) packets.



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Rules Policies (Chains)

Strings are sets of rules that we'll write to each table.

- **ACCEPT**
  Packets that meet the criteria are accepted, they continue their way through the stack,
- **DROP**
  Packets that meet the criteria are rejected, they are forgotten, and an ICMP message is not even sent.
- **LOG**
  It is a special target that allows us to trace by means of syslogs the packets that satisfy the criteria.
- **REJECT** (similar to DROP, but with an ICMP error message sent to the source of the rejected packet)
- **RETURN, REDIRECT, SNAT, DNAT, MASQUERADE**...

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Le DNAT | NAT Destination

- The PREROUTING chain allows you to change only the destination address but retains the source address.
- This is what we call doing DNAT (NAT destination).
- In this example, we will redirect traffic destined for the network "195.111.222.0" to the network "193.168.1.0".

iptables -t nat -A PREROUTING -d 195.111.222.0/24 -j DNAT --to-destination 193.168.1.0/24

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Le SNAT | NAT Source

- The POSTROUTING chain (after routing) allows you to change only the source address but keeps the destination address.
- This is what we call doing SNAT (NAT source).
- In this example, we will substitute the source address of outbound private traffic with one of the three addresses.

iptables -t nat -A POSTROUTING -s 193.168.1.0/24 -j SNAT --to-source 192.168.1.5-192.168.1.8

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IP Masquerade

- The principle is to use a source public IP address to hide all the IP addresses of the private network behind it
  - **iptables -t nat -A POSTROUTING -s 193.168.1.0/24 -j SNAT --to-source 192.168.1.199** is equivalent to **iptables -t nat -A POSTROUTING -s 193.168.1.0/24 -j MASQUERADE**
- You can use other options, such as choosing in relation to your network interface, in this case "eth0" and the port number, in this case "80".
  - **iptables -t nat -A POSTROUTING -o eth0 --dport 80 -j MASQUERADE**

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The state machine

- **NEW:** New connection
- **ESTABLISHED:** Connection already established
- **RELATED** : connection with another already established
- **INVALID**: The invalid packet

➔In general, they work with the --state match

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Creating a Rule

- iptables [-t *table*] commande [correspondance] [cible]

iptables(8) - Linux man page

Name

iptables - administration tool for IPv4 packet filtering and NAT

Synopsis

iptables [-t table] -[AD] chain rule-specification [options]
iptables [-t table] -I chain [rulenum] rule-specification [options]
iptables [-t table] -R chain rulenum rule-specification [options]
iptables [-t table] -D chain rulenum [options]
iptables [-t table] -[LFZ] [chain] [options]
iptables [-t table] -N chain
iptables [-t table] -X [chain]
iptables [-t table] -P chain target [options]
iptables [-t table] -E old-chain-name new-chain-name

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DDoS and Ping of Death Protection

- **iptables -A INPUT -p icmp --icmp-type echo-request -m limit --limit 1/s -j ACCEPT**
- **iptables -A INPUT -p icmp --icmp-type echo-request -j DROP**

  - **-A INPUT** → for adding a rule to the filter point (INPUT, OUTPUT, FORWARD, PREROUTING, POSTROUTING) which is also called a string.
  - **-p icmp** → Targets ICMP packets (used for ping).
  - **--icmp-type echo-request** → Filters only "ping" requests.
  - **-m limit --limit 1/s** → Allows a maximum of 1 ping per second.
  - **-j ACCEPT** → Accepts ping within the set limit.
  - **-j DROP** → Blocks excess pings

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Protection against ICMP (Flooding, Smurf Attack) attacks

- **iptables -A INPUT -p icmp -m limit --limit 1/s --limit-burst 3 -j ACCEPT**
- **iptables -A INPUT -p icmp -j DROP**

--limit 1/s --limit-burst 3 → Allows up to 3 ICMP packets bursting, then limits to 1 packet per second.
Deuxième règle -j DROP → Blocks any excess ICMP requests.

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Protection against SYN Flood (TCP-SYN Attack)

- iptables -A INPUT -p tcp --syn -m limit --limit 10/s --limit-burst 20 -j ACCEPT
- iptables -A INPUT -p tcp --syn -j DROP

--syn → Targets TCP SYN connection packets.
--limit 10/s --limit-burst 20 → Allows 20 simultaneous connections and then limits to 10 per second.
-j DROP→ Blocks excessive SYN connections.

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Block fragmented packets (used in some DDoS attacks)

- iptables -A INPUT -f -j DROP

-f → Targets fragmented packets.
-j DROP → Blocks these packets to prevent fragmentation-based attacks.

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Block packets with spoofed IP addresses

- iptables -A INPUT -s 10.0.0.0/8 -j DROP
- iptables -A INPUT -s 172.16.0.0/12 -j DROP
- iptables -A INPUT -s 192.168.0.0/16 -j DROP

-s → Specifies the source address.

IP Range (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) → Corresponds to private addresses that should not appear on the Internet.

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Block a specific IP (e.g., address of a detected attacker)

- iptables -A INPUT -s 192.168.1.100 -j DROP

-s 192.168.1.100 → Blocks all packets from this IP.

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FORWARD chain: Management of traffic that passes through the machine

- Allow only HTTP traffic (port 80) that traverses the machine
  - **sudo iptables -A FORWARD -p tcp --dport 80 -j ACCEPT**
  - **sudo iptables -A FORWARD -j DROP**

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FORWARD chain: Management of traffic that passes through the machine

- Block packets from a specific IP address that pass through the router
  - **sudo iptables -A FORWARD -s 192.168.126.100 -j DROP**

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FORWARD chain: Management of traffic that passes through the machine

- Limit SYN traffic traversing the router (vs. SYN Flood)
  - **sudo iptables -A FORWARD -p tcp --syn -m limit --limit 5/s --limit-burst 10 -j ACCEPT**
  - **sudo iptables -A FORWARD -p tcp --syn -j DROP**

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OUTPUT Chain: Managing Outbound Traffic from the Machine

- Block any ping (ICMP echo-request) leaving the machine
  - **sudo iptables -A OUTPUT -p icmp --icmp-type echo-request -j DROP**

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OUTPUT Chain: Managing Outbound Traffic from the Machine

- Limit outbound pings to 1 ping per second
  - **sudo iptables -A OUTPUT -p icmp --icmp-type echo-request -m limit --limit 1/s -j ACCEPT**
  - **sudo iptables -A OUTPUT -p icmp --icmp-type echo-request -j DROP**

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OUTPUT Chain: Managing Outbound Traffic from the Machine

- Allow only HTTPS traffic (port 443) for the local machine
  - **sudo iptables -A OUTPUT -p tcp --dport 443 -j ACCEPT**
  - **sudo iptables -A OUTPUT -j DROP**

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Block IP ranges (spoofing protection) on FORWARD and INPUT

- Anti-spoofing rule for private addresses that should not be routed over the Internet
  - **sudo iptables -A INPUT -s 10.0.0.0/8 -j DROP**
  - **sudo iptables -A INPUT -s 172.16.0.0/12 -j DROP**
  - **sudo iptables -A INPUT -s 192.168.0.0/16 -j DROP**

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Block Ftp/SSH & @MAC

- ***iptables -A INPUT -p tcp -m multiport --destination-port 21,23 -s ! 127.0.0.1 -j DROP***
  - closing the ports of the FTP servers and Telnet (the password circulates in clear text on the network) [these two applications must be replaced by SFTP and SSH]. With this rule, FTP and Telnet servers become unresponsive except for local loop tests (address 127.0.0.1).
- ***iptables -A INPUT -p tcp --syn -m limit --limit 10/s -j ACCEPT***
  - Limiting the number of Connection Open Requests (SYNs) to 10 per second to prevent the SYN flooding attack.
- ***iptables -A INPUT -p tcp -m mac --mac-source 00:02:B3:98:41:08 –j DROP***
  - Banned all TCP connections from the machine with MAC address 00:02:B3:98:41:08. All TCP segments are rejected.

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